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Automated feature selection of predictors in electronic medical records data

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Abstract

The use of Electronic Health Records (EHR) for translational research can be challenging due to difficulty in extracting accurate disease phenotype data. Historically, EHR algorithms for annotating phenotypes have been either rule-based or trained with billing codes and gold standard labels curated via labor intensive medical chart review. These simplistic algorithms tend to have unpredictable portability across institutions and low accuracy for many disease phenotypes due to imprecise billing codes. Recently, more sophisticated machine learning algorithms have been developed to improve the robustness and accuracy of EHR phenotyping algorithms. These algorithms are typically trained via supervised learning, relating gold standard labels to a wide range of candidate features including billing codes, procedure codes, medication prescriptions and relevant clinical concepts extracted from narrative notes via Natural Language Processing (NLP). However, due to the time intensiveness of gold standard labeling, the size of the training set is often insufficient to build a generalizable algorithm with the large number of candidate features extracted from EHR. To reduce the number of candidate predictors and in turn improve model performance, we present an automated feature selection method based entirely on unlabeled observations. The proposed method generates a comprehensive surrogate for the underlying phenotype with an unsupervised clustering of disease status based on several highly predictive features such as diagnosis codes and mentions of the disease in text fields available in the entire set of EHR data. A sparse regression model is then built with the estimated outcomes and remaining covariates to identify those features most informative of the phenotype of interest. Relying on the results of Li and Duan (1989), we demonstrate that variable selection for the underlying phenotype model can be achieved by fitting the surrogate-based model. We explore the performance of our methods in numerical simulations and present the results of a prediction model for Rheumatoid Arthritis (RA) built on a large EHR data mart from the Partners Health System consisting of billing codes and NLP terms. Empirical results suggest that our procedure reduces the number of gold-standard labels necessary for phenotyping thereby harnessing the automated power of EHR data and improving efficiency.

KEYWORDS
electronic medical records, feature selection, prediction accuracy, regularized regression, risk prediction

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1 | INTRODUCTION

To improve their translational potential, omics studies have been increasingly conducted on patients using their Electronic Health Record (EHR) data. EHRs contain a wealth of clinical data collected as part of routine care and those linked with biobanks provide a robust and potentially cost-effective approach to omics studies. These EHR biobank platforms allow for direct links between the rich phenotypic data from the EHR and the omics data generated from analyses of biobank samples (Kohane, 2011; Murphy et al., 2009). In contrast to the speed and ease of amassing biologic data through omics analyses, progress in accurately and efficiently extracting disease phenotype data has lagged behind and become a major barrier to fully realizing the potential of EHR for translational research.

Historically, the International Classification of Diseases Ninth Edition (ICD-9) billing code information is used to classify a subjects's true binary disease status, Y, for claims and EHR based research studies. However, ICD-9 codes often have low accuracy in predicting the true Y, which can lead to loss of power and potential bias for downstream association analyses. Recently, much research has been focused on developing EHR phenotyping algorithms that can accurately predict disease phenotypes, for example, (Liao, et al., 2010; Xia et al., 2013). These algorithms make use of a wide range of candidate features specified by domain experts. Examples include billing codes, procedure codes, medication prescriptions, laboratory results and mentions of relevant clinical terms extracted from narrative text of clinical notes using natural language processing (NLP) (Liao et al., 2015). Since the number of candidate features is typically not small, a substantial number of gold standard labels are often needed to ensure stable algorithm training. This limits the scalability of such an approach as the labeling process is highly labor intensive.

In this paper, we aim to improve the efficiency of phenotyping by developing an unsupervised procedure based entirely on unlabeled data for identifying a subset of the candidate feature vector $\mathbf{X}$ that is predictive of Y. Our procedure leverages surrogate variables, denoted by $\mathbf{S}$, that can predict Y with moderate precision. For example, if the phenotype is rheumatoid arthritis (RA), we can have reasonable confidence that two surrogate variables are highly correlated with RA status: (1) the total count of the ICD-9 billing code for RA, and (2) the count of NLP mentions of the term "rheumatoid arthritis." Unlike Y, the surrogate variables $\mathbf{S}$ can be automatically extracted for all patients in the database. Moreover, they are likely as close to the true phenotype as one can measure without chart review. The statistical problem of interest can therefore be cast into a measurement error problem where $\mathbf{S}$ can be viewed as a vector of "mis-measured" variables of Y. Most existing literature on measurement error for binary response focuses on a single binary surrogate and/or require some observations on Y (Brenner and Gefeller, 1993; Carroll et al., 2006; Gerlach and Stamey, 2007; Greenland, 2008; Magder and Hughes, 1997; Neuhaus, 1999). Misclassification bias is typically mitigated under the assumption of known misclassification probabilities or with a labeled dataset that enables quantification of the sensitivity and specificity of S. In the absence of any observations on Y, Magder and Hughes (1997) demonstrated that an expectation maximization algorithm can be used to obtain the maximum likelihood estimator for the outcome model as well as the accuracy of a binary S. However, in a finite sample, the scale of the regression parameters and the accuracy parameters cannot be well estimated due to the flatness of the likelihood function.

In the EHR setting, there are often multiple non-binary surrogates. Few feature selection procedures currently exist for the setting when Y is never observed but there are abundant observations on multiple surrogates $\mathbf{S}$. Within the EHR literature, Yu et al. (2015) proposed a simple surrogate based feature selection method using a marginal absolute rank correlation between S and each $\mathbf{X}$. Yu et al. (2016) and Agarwal et al. (2016) used a surrogate S as a "silver standard label" to perform feature selection with sparse logistic regression of S against $\mathbf{X}$ to infer about the relationship between Y and $\mathbf{X}$. However, no theoretical justifications were provided for these procedures and it is unclear under what conditions such procedures are valid.

To overcome these challenges, we propose a two-step unsupervised feature selection method by first turning $\mathbf{S}$ into an estimate of $P(Y = 1 \mid \mathbf{S})$, denoted by $\pi_{\mathbf{S}}$, and then performing a sparse regression of $\pi_{\mathbf{S}}$ against $\mathbf{X}$. Under the assumption that $\mathbf{S} \perp \mathbf{X} \mid Y$ and building on the work of Neuhaus (1999), we show that regression parameter from the fitted quasi-logistic model of $\pi_{\mathbf{S}}$ and $\mathbf{X}$ is consistent up to a scalar for the parameter under a generalized linear model (GLM) (McCullagh and Nelder, 1989) of $Y \mid \mathbf{X}$. This implies that variable selection for Y may be achieved using only the unlabeled EHR data. Consequently, the small number of observations that do have gold standard labels may be used to develop a more parsimonious model trained on the selected variables from the surrogate model. Additionally, we propose a robust resampling procedure to further improve the stability of our variable selection method for the purposes of prediction. Empirical results suggest that our approach can outperform models trained solely on labeled data and in turn has potential to reduce the manual chart review impeding EHR-based research.

The remainder of this paper is organized as follows. Section 2 describes the proposed feature selection procedure. Section 3.1 presents results from simulation studies that demonstrate the potential utility of our method in practice. We then illustrate our procedures in Section 3.2 with feature selection in the development of a phenotyping algorithm to identify rheumatoid arthritis cases using EHR data from the Partners Health System. Concluding remarks are provided in Section 4.

2 | METHODS

Letting Y denote the binary phenotype, our goal is to identify the subset of EHR features $\mathbf{X}_{p\times 1}$ that are predictive of Y leveraging surrogate features $\mathbf{S}_{q\times 1}$. The underlying data consist of N independent and identically distributed random vectors $\mathscr{F} = \{(Y_i, \mathbf{X}_i^\mathsf{T}, \mathbf{S}_i^\mathsf{T})^\mathsf{T}, i = 1, \ldots, N\}$, but due to the difficulty in obtaining information on Y only $\mathscr{D} = \{\mathbf{W}_i = (\mathbf{X}_i^\mathsf{T}, \mathbf{S}_i^\mathsf{T})^\mathsf{T}, i = 1, \ldots, N\}$ is observed. Throughout, we assume that $\mathbf{S}$ depends on $\mathbf{X}$ only through Y, that is, (Carroll et al., 2006).

$$\mathbf{S} \perp \mathbf{X} | Y \tag{A.1}$$

Further, Y is assumed to follow a GLM

$$P(Y = 1|\mathbf{X}) = g(\alpha_0 + \mathbf{X}^\mathsf{T}\boldsymbol{\beta}_0) = g(\vec{\mathbf{X}}^\mathsf{T}\boldsymbol{\theta}_0), \\ \text{with } \vec{\mathbf{X}} = (1, \mathbf{X}^\mathsf{T})^\mathsf{T}, \quad \boldsymbol{\theta}_0 = (\alpha_0, \boldsymbol{\beta}_0^\mathsf{T})^\mathsf{T}, \tag{A.2}$$

where $g(\cdot)$ is a known smooth link function. We define the support of $\boldsymbol{\beta}_0$ as $\mathcal{A} = \{j | \beta_{0j} \neq 0\}$, $\mathcal{A}^c = \{j | \beta_{0j} = 0\}$ and $\mathbf{X}_{\mathcal{B}}$ the subvector of $\mathbf{X}$ corresponding to $\mathcal{B}$ for any set $\mathcal{B}$. We next detail our automated feature selection procedure for identifying the active set $\mathcal{A}$.

2.1 | Unsupervised feature selection procedure

Step I: Clustering. Our proposed feature selection procedure starts with a first step of estimating the predicted probability of $Y = 1$ given $\mathbf{S}$, denoted by $\pi_{\mathbf{S}} = P(Y = 1 \mid \mathbf{S})$. We impose a *working* parametric mixture model of $\mathbf{S}$ as

$$\mathbf{S} \sim \tau f_{\boldsymbol{\Theta}_1}(\mathbf{s}) + (1-\tau) f_{\boldsymbol{\Theta}_0}(\mathbf{s}) \tag{1}$$

where $f_{\boldsymbol{\Theta}_y}(\cdot)$ are distribution functions of $\mathbf{S} \mid Y = y$ specified up to a vector of unknown parameters $\boldsymbol{\Theta}_y$ for $y = 0, 1$ and $\tau = P(Y = 1)$ is the unknown prevalence. Commonly used mixture models include the multivariate normal mixture and the poisson mixture (Titterington et al., 1985; Fraley and Raftery, 2002; Leisch, 2004; McLachlan and Peel, 2004).

To estimate the model parameters $\boldsymbol{\Theta}_\bullet = (\boldsymbol{\Theta}_1^\mathsf{T}, \boldsymbol{\Theta}_0^\mathsf{T}, \tau)^\mathsf{T}$, we use the standard maximum likelihood estimator (MLE) obtained via the Expectation-Maximization algorithm (Dempster et al., 1997). Letting $\widehat{\boldsymbol{\Theta}}_\bullet = (\widehat{\boldsymbol{\Theta}}_1^\mathsf{T}, \widehat{\boldsymbol{\Theta}}_0^\mathsf{T}, \hat{\tau})^\mathsf{T}$ denote the maximizer of $\widehat{\mathcal{L}}_{\text{mix}}(\boldsymbol{\Theta}_\bullet) = N^{-1}\sum_{i=1}^N \mathcal{L}_{mix}(\mathbf{S}_i, \boldsymbol{\Theta}_\bullet)$, where $\mathcal{L}_{mix}(\mathbf{S}_i, \boldsymbol{\Theta}_\bullet)$ is the log-likelihood contribution from the ith subject, we then approximate $\pi_{\mathbf{S}}$ as

$$\hat{\pi}_{\mathbf{S}} = \Pi_{\mathbf{S}}(\widehat{\boldsymbol{\Theta}}_\bullet) \quad \text{where} \quad \Pi_{\mathbf{S}}(\boldsymbol{\Theta}_\bullet) = \frac{\tau f_{\boldsymbol{\Theta}_1}(\mathbf{S})}{\tau f_{\boldsymbol{\Theta}_1}(\mathbf{S}) + (1-\tau) f_{\boldsymbol{\Theta}_0}(\mathbf{S})}. \tag{2}$$

We assume that regardless of the adequacy of (1), $E\{\mathcal{L}_{mix}(\mathbf{S}_i, \boldsymbol{\Theta}_\bullet)\}$ has a unique maximizer $\bar{\boldsymbol{\Theta}}_\bullet = \text{argmax}_{\boldsymbol{\Theta}_\bullet} E\{\mathcal{L}_{mix}(\mathbf{S}_i, \boldsymbol{\Theta}_\bullet)\}$ up to a permutation of components. Then under mild regularity conditions, one may show that $\widehat{\boldsymbol{\Theta}}_\bullet$ and $\hat{\pi}_{\mathbf{S}}$ are $\sqrt{N}$-consistent estimators of $\bar{\boldsymbol{\Theta}}_\bullet$ and $\bar{\pi}_{\mathbf{S}} = \Pi_{\mathbf{S}}(\bar{\boldsymbol{\Theta}})$, respectively. When model (1) is correctly specified, $\bar{\boldsymbol{\Theta}} = \boldsymbol{\Theta}_\bullet$ and $\bar{\pi}_{\mathbf{S}} = \pi_{\mathbf{S}}$.

Step II: Regularized Estimation. In the second step, we fit a penalized quasi-logistic regression of $\hat{\pi}_{\mathbf{S}_i}$ against $\mathbf{X}_i$. Specifically, let $\hat{\boldsymbol{\theta}} = (\hat{\alpha}, \hat{\boldsymbol{\beta}}^\mathsf{T})^\mathsf{T} = \text{argmin}_{\boldsymbol{\theta}} \hat{\mathcal{L}}(\boldsymbol{\theta})$ where

$$\hat{\mathcal{L}}(\boldsymbol{\theta}) = N^{-1}\sum_{i=1}^N \ell(\boldsymbol{\theta}^\mathsf{T}\vec{\mathbf{X}}_i, \hat{\pi}_{\mathbf{S}_i}) + \lambda_N \sum_{j=1}^p |\beta_j| / |\tilde{\beta}_j|, \tag{3}$$

$\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^\mathsf{T})^\mathsf{T}$, $\ell(x_i, \pi_i)$ is the negative log-likelihood contribution from the ith subject, $\tilde{\boldsymbol{\theta}} = (\tilde{\alpha}, \tilde{\boldsymbol{\beta}}^\mathsf{T})^\mathsf{T}$ is the minimizer of $\tilde{\mathcal{L}}(\boldsymbol{\theta}) = N^{-1}\sum_{i=1}^N \ell(\boldsymbol{\theta}^\mathsf{T}\vec{\mathbf{X}}_i, \hat{\pi}_{\mathbf{S}_i})$, and λ_N is such that $N^{1/2}\lambda_N \to \infty$ and $\lambda_N \to 0$ as $N \to \infty$. Note that $\lambda_N \sum_{j=1}^p |\beta_j|/|\tilde{\beta}_j|$ is the adaptive least absolute shrinkage and selection operator (ALASSO) penalty. We then consider the features in the active set $\hat{\mathcal{A}} = \{j : \hat{\beta}_j \neq 0\}$ to be those predictive of Y.

In practice, solving (3) may be computationally intensive due to the magnitude of N. We, therefore, employ a quadratic approximation to Wang and Leng(3) as in Wang and Leng (2007) and instead minimize

$$\|\tilde{Y} - \boldsymbol{\theta}^\mathsf{T}\tilde{\mathbb{X}}\|_2^2 + \lambda_N \sum_{j=1}^p |\beta_j|/|\tilde{\beta}_j|$$

where $\tilde{Y} = \tilde{\boldsymbol{\theta}}^\mathsf{T}\tilde{\mathbb{X}}$, $\tilde{\mathbb{X}}$ is the symmetric half matrix of $\tilde{\mathbb{I}} = N^{-1}\sum_{i=1}^N \ddot{\ell}(\tilde{\boldsymbol{\theta}}^\mathsf{T}\vec{\mathbf{X}}_i, \hat{\pi}_{\mathbf{S}_i})$ such that $\tilde{\mathbb{I}} = \tilde{\mathbb{X}}^\mathsf{T}\tilde{\mathbb{X}}$, and $\ddot{\ell}(x, \pi) = \partial^2\ell(x, \pi)/\partial x^2$.

To motivate this procedure, we note that under assumptions (A.1) and (A.2), the density function of $\bar{\pi}_{\mathbf{S}}$ given $\mathbf{X}$ can be written as

$$f(t|\mathbf{X}) = \dot{G}_1(t)P(Y = 1|\mathbf{X}) + \dot{G}_0(t)P(Y = 0|\mathbf{X}) = \dot{G}_1(t)g(\vec{\mathbf{X}}^\mathsf{T}\boldsymbol{\theta}_0) \\ + \dot{G}_0(t)\{1 - g(\vec{\mathbf{X}}^\mathsf{T}\boldsymbol{\theta}_0)\}$$

where $\dot{G}_y(t) = dG_y(t)/dt$ and $G_y(t) = P(\bar{\pi}_{\mathbf{S}} \leq t | Y = y)$. Thus, $\bar{\pi}_{\mathbf{S}} \mid \mathbf{X}$ follows a single index model (SIM) in that $\bar{\pi}_{\mathbf{S}}$ relates to $\mathbf{X}$ only through $\mathbf{X}^\mathsf{T}\boldsymbol{\beta}_0$. It is also interesting to note that

$$E(\bar{\pi}_{\mathbf{S}}|\mathbf{X}) = \int_0^1 P(\bar{\pi}_{\mathbf{S}} > t|\mathbf{X})dt = \mu_0 + g(\boldsymbol{\theta}_0^{\mathsf{T}}\vec{\mathbf{X}})[\mu_1 - \mu_0] \quad (4)$$

where $\mu_y = E(\bar{\pi}_{\mathbf{S}}|Y = y)$. This result coincides with that of Neuhaus (1999) when $\bar{\pi}_{\mathbf{S}} = S$ and implies that the model for $P(Y = 1|\mathbf{X})$ fit with $\bar{\pi}_{\mathbf{S}}$ instead of Y is equivalent to a misspecification of the link function. We can also see from (4) that if μ_1 is close to 1 and μ_0 is close to 0, which is what we would expect for a good surrogate outcome following approximately the mixture model (1), $E(\bar{\pi}_{\mathbf{S}}|\mathbf{X})$ is a close approximation to the true conditional risk function. On the other hand, the fact that $\bar{\pi}_{\mathbf{S}} \mid \mathbf{X}$ follows a SIM that only depends on $\mathbf{X}$ through $\boldsymbol{\beta}_0^{\mathsf{T}}\mathbf{X}$ may be leveraged even if the mixture model fails to hold. That is, provided that $E(\mathbf{b}^{\mathsf{T}}\mathbf{X} \mid \mathbf{X}^{\mathsf{T}}\boldsymbol{\beta}_0)$ is linear in $\mathbf{X}^{\mathsf{T}}\boldsymbol{\beta}_0$ for all linear combinations $\mathbf{b}^{\mathsf{T}}\mathbf{X}$, $\bar{\pi}_{\mathbf{S}}$ only needs to be predictive of Y for our approach to recover $\boldsymbol{\beta}_0$ proportionally due to the results of Li and Duan (1989) and the convexity of the loss in (3). Thus, $\widehat{\boldsymbol{\theta}}$ may be used for feature selection. In the Appendix in the Supplementary Materials, we outline justifications for the validity of the feature selection procedure and demonstrate that $P(\widehat{\mathcal{A}} = \mathcal{A}) \to 1$ as $N \to \infty$.

2.2 | Variable selection via resampling

Although $P(\widehat{\mathcal{A}} = \mathcal{A}) \to 1$ as $N \to \infty$, phenotyping algorithms trained with $\mathbf{X}_{\widehat{\mathcal{A}}}$ do not perform as well as algorithms trained using $\mathbf{X}_{\mathcal{A}}$ due to the uncertainty in $\widehat{\mathcal{A}}$, particularly when the number of gold standard labels is relatively small and the inclusion of weak or null signals can substantially impact out of sample prediction performance. To improve the stability in the feature selection, we propose the use of a resampling method to obtain an estimate of $P(\widehat{\beta}_j = 0)$, denoted $\widehat{\rho}_{0j}$, and select features based on $\widehat{\rho}_{0j}$. However, when N is large, standard resampling procedures may be computationally intensive. In recent years, bootstrap variants such as the m out of n bootstrap (Bickel, 1997) and subsampling (Politis and Wolf, 1999) have been introduced to address settings where the bootstrap fails asymptotically. Though not their initial purpose, these procedures also yield computational benefits as they perform the repeated estimation on more manageable subsets of the original data (Bickel and Sakov, 2002). Moreover, when N is sufficiently large as is typically the case in the EHR setting, the size of the subsample, denoted by N_b, may be chosen so that computational gains are achieved without loss of performance.

For the mth subsample, indexed by $\mathcal{R}_m$, we obtain a resampled counterpart $\widehat{\boldsymbol{\theta}}^{(m)} = (\widehat{\alpha}^{(m)}, \widehat{\boldsymbol{\beta}}^{(m)\mathsf{T}})^{\mathsf{T}}$ of $\widehat{\boldsymbol{\theta}}$ as the minimizer of

$$\widehat{\mathcal{L}}^{(m)}(\boldsymbol{\theta}) = N_b^{-1} \sum_{i \in \mathcal{R}_m} \ell(\boldsymbol{\theta}^{\mathsf{T}}\vec{\mathbf{X}}_i, \widehat{\pi}_{\mathbf{S}_i}) + \lambda_N^{(m)} \sum_{j=1}^{p} |\beta_j| / |\widetilde{\beta}_j^{(m)}|$$

Note that we do not need to re-restimate $\widehat{\pi}_{\mathbf{S}_i}$ in each resample due to the size of N. Then, based on a large number M of subsamples, we obtain

$$\widehat{\rho}_{0j} = M^{-1} \sum_{m=1}^{M} I(\widehat{\beta}_j^{(m)} = 0)$$

Variables are correspondingly selected based on a cutoff, ρ_{cut}, so that we select X_j when $\widehat{\rho}_{0j} < \rho_{cut}$. Though any threshold $\rho_{cut} \in (0, 1)$ will guarantee that

$$P\left\{ M^{-1} \sum_{m=1}^{M} I(\widehat{\beta}_j^{(m)} = 0) < \rho_{cut} \right\} \to I(j \in \mathcal{A})$$

conditional on the observed data due to the oracle properties of $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\beta}}^{(m)}$, we have found that the standard choice of 0.5 works well in practice.

3 | NUMERICAL STUDIES

3.1 | Simulation

We examine the performance of our method on simulated data sets under three settings. In setting 1, we consider the scenario in which all assumptions hold, that is, $E(\mathbf{b}^{\mathsf{T}}\mathbf{X} \mid \mathbf{X}^{\mathsf{T}}\boldsymbol{\beta}_0)$ is linear in $\mathbf{X}^{\mathsf{T}}\boldsymbol{\beta}_0$ and the logistic regression model for Y is correctly specified. In setting 2, we explore the robustness of our proposed procedures when both of these assumptions are violated but $S \perp \mathbf{X} \mid Y$. In setting 3, we consider the scenario where $\mathbf{S}$ is not conditionally independent of $\mathbf{X}$ given Y. In all scenarios, we simulated $N = 5000$ subjects with binary outcome Y with prevalence 0.3 dependent on a varying number of covariates $\mathbf{X}$ and two surrogate markers $\mathbf{S}$. In Settings 1 and 3, we generated the covariate vectors as

$$\mathbf{X}_i \sim \text{MVN}(0, \boldsymbol{\Sigma}^{\mathbf{x}}) + \boldsymbol{\mu}_{y_i}^{\mathbf{x}} \quad \boldsymbol{\mu}_{y_i}^{\mathbf{x}} = y_i \boldsymbol{\Sigma}^{\mathbf{x}} \boldsymbol{\beta}_0,$$

with $\Sigma_{kk}^{\mathbf{x}} = 1$, and $\Sigma_{kj}^{\mathbf{x}} = \varrho^{|k-j|}$. In Setting 2, we generated covariates analogously to Settings 1 and 3 and then applied the transformation $\log(\lfloor \exp(\mathbf{X}) \rceil + 1)$ such that the elliptical distributional assumption does not hold for $\mathbf{X}$, where $\lfloor \cdot \rceil$ denotes the nearest integer function. To mimic EHR data, we generated $\mathbf{S}$ as count variables with $\mathbf{S} = \log\{\lfloor \exp(\mathbf{S}^0) \rceil + 1\}$, where

$$\text{Settings 1 \& 2:} \quad \mathbf{S}_i^0 \sim \text{MVN}(0, \boldsymbol{\Sigma}^s) + \boldsymbol{\mu}^{\mathbf{s}} + y_i \boldsymbol{\Delta}_0^{\mathbf{s}}$$

$$\text{Setting 3:} \quad \mathbf{S}_i^0 \sim \text{MVN}(0, \boldsymbol{\Sigma}^s) + \boldsymbol{\mu}^{\mathbf{s}} + y_i \boldsymbol{\Delta}_0^{\mathbf{s}} + [X_1, X_3],$$

$\boldsymbol{\mu}^{\mathbf{s}} = [-1, -2]^{\mathsf{T}}$, $\Sigma_{11}^s = 0.65$, $\Sigma_{12}^s = 7.3$, and $\Sigma_{22}^s = 0.65$. Under Setting 1, we have the correct specification of $P(Y = 1 \mid \mathbf{X}) = 1/(1 + e^{-(\alpha_0 + \boldsymbol{\beta}_0^{\mathsf{T}}\mathbf{X})})$. In all scenarios, we considered $\varrho = 0.5$, $p = 50$ or 100, and the number of labeled samples available for final algorithm training after feature selection set to $n = 100$ or 200. We also considered two sets of $\boldsymbol{\beta}_0$ and $\boldsymbol{\Delta}_0^{\mathbf{s}}$:

$$\textit{strong signal with } \boldsymbol{\beta}_0 = [1.2, -1.2, 0.5, -0.3, 0.3, 0.1, 0.1, \mathbf{0}_{(p-7)\times 1}^{\mathsf{T}}]^{\mathsf{T}}, \boldsymbol{\Delta}_0^{\mathbf{s}} = [0.75, 3]^{\mathsf{T}}$$
$$\textit{and weak signal with } \boldsymbol{\beta}_0 = [0.9, -0.9, 0.5, -0.3, 0.3, 0.1, 0.1, \mathbf{0}_{(p-7)\times 1}^{\mathsf{T}}]^{\mathsf{T}}, \boldsymbol{\Delta}_0^{\mathbf{s}} = [0.8, 2.2]^{\mathsf{T}}.$$

For the strong signal, the AUCs and standard deviations of $\{S_1, S_2\}$ are, respectively, $\{0.701, 0.752\}$ and $\{0.43, 1.41\}$ in Settings 1 and 2; $\{0.764, 0.749\}$ and $\{0.71, 1.53\}$ in Setting 3. For the weak signal, the AUCs and standard deviations of $\{S_1, S_2\}$ are, respectively, $\{0.714, 0.683\}$ and $\{0.44, 1.24\}$ in Settings 1 and 2; $\{0.747, 0.691\}$ and $\{0.69, 1.38\}$ in Setting 3. For conciseness, we report the strong signal setting only and include the results for the weak signal in the Supplementary Materials as both scenarios follow similar patterns. Throughout, we utilized R package `mclust` (Fraley et al., 2012) under the Gaussian mixture model (GMM) (Fraley and Raftery, 2002) to estimate $\pi_{\mathbf{S}}$, where the optimal model is chosen according to Bayesian Information Criterion. The ALASSO penalized regression was fit with penalty parameter λ_N selected using a modified BIC as in Minnier et al. (2011). For the resampling procedure, we used $M = 500$ with $N_b = 2000$ in each setting and $\rho_{cut} = 0.5$.

As a benchmark, we performed various supervised learning methods by fitting (I) an ALASSO penalized logistic regression of Y against $\mathbf{X}$ and $\mathbf{S}$ for a training sample of size $n = 100$ and 200 (L_{100} and L_{200}); and (II) a two-step supervised method where the first step involves the regression of Y on $\mathbf{X}$ followed by the regression of Y on the selected features from step 1 as well as $\mathbf{S}$ (L_{100}^{2step}, L_{200}^{2step}). For each dataset, we also trained final phenotyping algorithms by fitting a penalized logistic regression model to a training set of size $n = 100$ or 200 using $\mathbf{S}$ plus features selected by (i) our automated clustering based feature selection procedure (AutoClust) or (ii) our automated feature selection procedure with resampling (AutoClust$_R$). For the algorithms trained with the labeled examples, the initial estimate for ALASSO was obtained from ridge regression with ridge parameter equal to p/N.

For comparison, we evaluated models trained with $\mathbf{S}$ plus features selected from (i) ALASSO penalized linear regression with $S_1 + S_2$ as the outcome and $\mathbf{X}$ as the covariates (PenReg$_{S_1+S_2}$); (ii) a multivariate penalized regression (PenReg$_{\mathbf{S}}$) method by selecting features according to the support of $|\hat{\boldsymbol{\gamma}}_1| + |\hat{\boldsymbol{\gamma}}_2|$, where $\{\hat{\alpha}_1, \hat{\boldsymbol{\gamma}}_1, \hat{\alpha}_2, \hat{\boldsymbol{\gamma}}_2\}$ is the minimizer of $\sum_{k=1}^{2}\sum_{i=1}^{N}\|S_{ki} - \alpha_k - \boldsymbol{\gamma}_k^{\mathsf{T}}\mathbf{X}_i\|^2 + \lambda\sum_{j=1}^{p}\sqrt{\gamma_{1j}^2 + \gamma_{2j}^2}$, $\boldsymbol{\gamma}_{k}$, is the k^{th} row and $\boldsymbol{\gamma}_{\cdot j}$ is the jth column of $\boldsymbol{\gamma}$; (iii) the rank correlation method of Yu et al. (2015) based on $S_1 + S_2$ (RankCor$_{S_1+S_2}$) with an absolute rank correlation threshold of 0.15; and (iv) the extreme sampling method of Yu et al. (2016) (Extreme). For the Extreme method, we obtained the silver standard labels, $\{S_C^*, S_1^*, S_2^*\}$, as

$$S_C^* = \begin{cases} 0 & if (S_1^* + S_2^*)/2 < 0.5 \\ 0.5 & if (S_1^* + S_2^*)/2 = 0.5 \\ 1 & if (S_1^* + S_2^*)/2 > 0.5, \end{cases} \quad \text{and}$$

$$S_i^* = \begin{cases} 0 & if\, S_i \le L_{S_i} \\ 0.5 & if\, L_{S_i} < S_i \le U_{S_i} \\ 1 & if\, S_i > U_{S_i} \end{cases} \quad \text{for i = 1,2,}$$

with L_{S_i} and U_{S_i} set to the 2.5 and 97.5 percentiles of S_i. For the three choices of S_j^*, we sampled M_j^* observations with $S_j^* = 1$ and M_j^* observations with $S_j^* = 0$ where $M_j^* = \min(0.8n_{S_j^*}^1, 0.8n_{S_j^*}^0, 400)$ and $n_{S_j^*}^1$ and $n_{S_j^*}^0$ are the numbers of observations with $S_j^* = 1$ and 0 respectively. We repeatedly sampled observations from these two extremes 200 times, trained penalized logistic regression models based on S_j^*, and selected informative features as those that were nonzero at least 50% of time, averaged over the repeated fittings and the three choices of S_j^*. We also considered using the unsupervised method of Agarwal et al. (2016) for feature selection by regressing the silver standard label $I(S_1 + S_2 \ge 1)$ on $\mathbf{X}$ (Agarwal$_{S_1+S_2\ge 1}$). We record the prediction performance of the algorithm with the area under the receiver operating characteristic curve (AUC) estimated with an independent validation set with 5000 samples. All estimates are based on Monte Carlo simulations with 500 replicates.

3.1.1 | Setting 1: Correct model specifications

We first summarize the variable selection performance of each method with respect to the proportion of times each variable is selected for the strong signal setting in Table 1. For conciseness, we averaged across all zero-valued coefficients in each model as results were similar. In general, we see that both AutoClust and AutoClust$_R$ perform well in variable selection with high probability of selecting stronger signals and near zero probability of selecting null features. Incorporating uncertainty, the AutoClust$_R$ approach tends to select a sparser model compared to AutoClust, which is desirable when the number of labels for training is small. The surrogate based penalized regression methods, including PenReg$_{S_1+S_2}$, PenReg$_{\mathbf{S}}$, and Agarwal$_{S_1+S_2\ge 1}$, tend to include null features while the RankCor$_{S_1+S_2}$ and the Extreme methods tend to

TABLE 1 Percent of times each feature was selected over 500 replications as well as the average AUC attained for final algorithm training with 100 or 200 labeled examples (AUC_{100}, AUC_{200}) when the covariate distribution is elliptically symmetric for the strong signal based on (i) supervised training with all features and 100 or 200 labeled samples (L_{100}, L_{200}) or with features selected via the two-step approach (L^{2step}_{100}, L^{2step}_{200}) (ii) the automated feature selection method with and without resampling (AutoClust, $AutoClust_R$), (iii) a penalized regression with $S_1 + S_2$ or $\mathbf{S}$ as the outcome ($PenReg_{S_1+S_2}$, $PenReg_{\mathbf{S}}$), (iii) the rank correlation method based on $S_1 + S_2$ ($RCor_{S_1+S_2}$) (iv) the extreme sampling method (Extreme) and (iv) the method of Agarwal et al. (2016) based on $S_1 + S_2 \geq 1$ as the silver standard label ($Agarwal_{S_1+S_2\geq 1}$).

(a) $p = 50$										
Method	1.2	−1.2	0.5	−0.3	0.3	0.1	0.1	0	AUC_{100}	AUC_{200}
L_{100}	85	46	3	6	14	11	9	3	79.9	
L^{2step}_{100}	88	59	4	5	19	13	9	4	81.1	
L_{200}	100	99	26	12	46	32	28	10		84.5
L^{2step}_{200}	100	99	32	12	47	33	27	11		84.3
AutoClust	100	100	91	43	66	28	29	12	84.2	86.3
$AutoClust_R$	100	100	74	15	42	12	10	2	85.3	86.6
$PenReg_{S_1+S_2}$	100	100	100	96	97	65	66	45	81.9	85.4
$PenReg_{\mathbf{S}}$	100	100	99	65	96	79	72	34	82.4	85.6
$RankCor_{S_1+S_2}$	36	0	0	0	0	0	0	0	79.7	80.2
Extreme	100	95	1	0	6	4	1	0	85.3	86.2
$Agarwal_{S_1+S_2\geq 1}$	100	100	99	66	85	38	38	14	83.9	86.2
(b) $p = 100$										
Method	1.2	−1.2	0.5	−0.3	0.3	0.1	0.1	0	AUC_{100}	AUC_{200}
L_{100}	58	13	2	0	4	4	3	0	76.5	
L^{2step}_{100}	47	8	1	0	2	4	2	0	79.6	
L_{200}	98	84	6	3	25	20	13	3		82.5
L^{2step}_{200}	100	92	8	3	31	23	15	3		83.9
AutoClust	100	100	90	39	65	31	24	11	83.5	85.8
$AutoClust_R$	100	100	68	12	34	11	6	2	85.2	86.5
$PenReg_{S_1+S_2}$	100	100	100	96	99	67	62	46	79.6	84.4
$PenReg_{\mathbf{S}}$	100	100	97	42	95	78	61	20	82.1	85.4
$RankCor_{S_1+S_2}$	36	0	0	0	0	0	0	0	79.6	80.2
Extreme	98	58	0	0	3	0	0	0	84.1	84.8
$Agarwal_{S_1+S_2\geq 1}$	100	100	98	59	81	35	32	11	83.4	85.9

under select signals. Though the rank correlation method may perform well in some settings, it is sensitive to the choice of threshold and performs quite poorly in this scenario. The supervised approaches generally perform substantially worse than the unsupervised counterparts.

The performance of the feature selection also translates into the classification performance of the final algorithm. As also shown in Table 1, model accuracy is substantially improved by AutoClust and $AutoClust_R$ when compared with the supervised methods. For instance, for the setting with the strong signal and $p = 100$, the average AUC was 0.765 when directly training a supervised algorithm with $n = 100$ labels, 0.835 and 0.852, respectively, when training using features selected by AutoClust and $AutoClust_R$. The existing unsupervised methods that tend to omit true signals, including the $RankCor_{S_1+S_2}$ and Extreme methods, have lower AUCs of 0.796 and 0.841. On the other hand, $PenReg_{\mathbf{S}}$, $PenReg_{S_1+S_2}$, and Agarwal et al. (2016) have AUCs ranging from 0.796 to 0.834 as a result of overfitting. The performance of these procedures may be similarly improved with resampling. For example, in the setting with $n = 100$ and $p = 50$, $PenReg_{\mathbf{S}}$ with subsampling achieves an AUC of 0.845 while $AutoClust_R$ has an AUC of 0.853. With $p = 100$, $PenReg_{\mathbf{S}}$ with subsampling achieves an AUC of 0.851, attaining nearly identical predicative performance as $AutoClust_R$.

3.1.2 | Setting 2: Distribution of X not elliptically symmetric

Next, we examine the robustness of our procedure when the distribution of $\mathbf{X}$ is not elliptically symmetric and the corresponding GLM for Y is mis-specified. As it is unclear how to define the positivity rate in this setting since sparsity is not necessarily maintained under mis-specification, we present in Table 2 the average model sizes and across simulations. We observe similar patterns as in Setting 1, finding that the supervised approach results in an overly sparse model with relatively poor prediction performance, particularly when $p = 100$. The $PenReg_{S_1+S_2}$ and $PenReg_{\mathbf{S}}$ methods result in substantially larger models while the rank correlation and Extreme method based methods result in overly sparse models. The $Agarwal_{S_1+S_2\geq 1}$ yielded a model size comparable to AutoClust, but larger than that of $AutoClust_R$. The

TABLE 2 Model Sizes and AUCs for final algorithm training with (a) $p = 100$ or (b) $p = 200$ labeled samples (AUC_{100}, AUC_{200}) for the strong signal when $\mathbf{X}$ is not elliptically symmetric based on (i) supervised training with all features and 100 or 200 labeled samples (L_{100}, L_{200}) or with features selected via the two-step approach ($\text{L}_{100}^{2\text{step}}$, $\text{L}_{200}^{2\text{step}}$) (ii) the automated feature selection method with and without resampling (AutoClust, AutoClust_R), (iii) a penalized regression with $S_1 + S_2$ or $\mathbf{S}$ as the outcome ($\text{PenReg}_{S_1+S_2}$, $\text{PenReg}_\mathbf{S}$), (iii) the rank correlation method based on $S_1 + S_2$ ($\text{RCor}_{S_1+S_2}$) (iv) the extreme sampling method (Extreme) and (iv) the method of Agarwal et al. (2016) based on $S_1 + S_2 \geq 1$ as the silver standard label ($\text{Agarwal}_{S_1+S_2\geq 1}$).

(a) $p = 50$			
Method	Model Size	AUC_{100}	AUC_{200}
L_{100}	1	77.8	
$\text{L}_{100}^{2\text{step}}$	1.1	79.8	
L_{200}	4.7		82.8
$\text{L}_{200}^{2\text{step}}$	5.4		83.4
AutoClust	10.1	82.8	85
AutoClust_R	4.3	84	85.4
$\text{PenReg}_{S_1+S_2}$	26	80.3	84.3
$\text{PenReg}_\mathbf{S}$	19.6	81.3	84.6
$\text{RankCor}_{S_1+S_2}$	0.3	79.3	79.9
Extreme	1.3	82.5	83.3
$\text{Agarwal}_{S_1+S_2\geq 1}$	11	82.6	85
(b) $p = 100$			
Method	Model Size	AUC_{100}	AUC_{200}
L_{100}	0.5	75.6	
$\text{L}_{100}^{2\text{step}}$	0.2	78.7	
L_{200}	1.9		80
$\text{L}_{200}^{2\text{step}}$	2.4		82.6
AutoClust	14.9	82	84.7
AutoClust_R	4.6	83.9	85.3
$\text{PenReg}_{S_1+S_2}$	50.3	77.8	82.6
$\text{PenReg}_\mathbf{S}$	22	80.9	84.3
$\text{RankCor}_{S_1+S_2}$	0.3	79.3	79.8
Extreme	0.8	81	81.6
$\text{Agarwal}_{S_1+S_2\geq 1}$	15.1	82	84.8

trends in AUC are also similar to Setting 1 where we again find the AutoClust_R provide substantial increases in predictive performance. For instance, for $p = 100$ the average AUC was 0.756 when directly training a supervised algorithm with all features and 100 labeled examples, 0.839 when training using features selected by AutoClust_R, and ranged from 0.778 to 0.820 for other unsupervised feature selection methods.

3.1.3 | Setting 3: $\mathbf{S} \not\perp\!\!\!\perp \mathbf{X} \mid Y$

Lastly, we present the performance of our method in the setting where assumption (A.1) is violated so that $\mathbf{S}$ is not conditionally independent of $\mathbf{X}$ given Y. The patterns are analogous to those observed in Settings 1 and 2 where we found that AutoClust and AutoClust_R substantially improve both variable selection and predictive performance relative to the supervised method. Relative to the existing methods, AutoClust and AutoClust_R tend to provide a balance between accurate variable selection and high predictive performance of the final algorithm. The final algorithm trained with features selected from AutoClust_R again attains the highest AUC compared to all other methods. It is encouraging to observe such improvements in both feature selection and prediction performance from our proposed methods compared to the standard supervised approach even under Settings 2 and 3. Our results indicate that we may achieve similar or higher prediction accuracy as the standard supervised procedure if we train algorithms using features selected via AutoClust_R using half as many labeled examples. This suggests that the laborious chart review involved in EHR phenotyping could be substantially reduced through the use of our method.

3.2 | Data analysis

We applied our feature selection methods to an EHR cohort from Partners Healthcare System to identify patients with rheumatoid arthritis (RA), the most common autoimmune joint disease worldwide affecting 1% of the population. The initial EHR cohort consists of 46 111 potential RA subjects defined as those with at least one ICD-9 code for RA and related diseases or subjects who had received a common RA diagnosis test (Liao, et al., 2010). A subset of 435 subjects was randomly selected to have their true RA status confirmed by a team of rheumatologists via medical chart review. We use these gold standard labels to validate our feature selection procedures.

The candidate features were assembled by parsing online knowledge sources (Wikipedia, Medscape, Merck Manual, NLM, and Mayo Clinic) via named entity recognition and mapped to the Unified Medical Language System as described in Yu et al. (2016). A total of 144 NLP features that appear in at least 3 of the 5 sources were used as candidate variables. We removed all variables with less than 10% non-zero values resulting in a total of $p = 79$ features for consideration. We let S_1 be the number of recorded RA ICD-9 codes, S_2 be the counts of NLP mentions of RA in patient records, and the remaining features be $\mathbf{X}$. All count features were taken as $x \mapsto \log(1 + x)$ to stabilize model fitting and standardized to have unit variance. Similar to our simulation studies, we randomly sampled $N = 5000$ unlabeled observations for feature selection. For the AutoClust_R method, we considered $N_b = 2000$ subsamples and let $L_{Si} = 0$ and $U_{S_i} = 10$ for the Extreme method as in Yu et al. (2016). For each method, we validated performance by training phenotyping algorithms with selected features using $n = 100$ and 200 labels. We also included the total number of notes in each patient's record to adjust for patient healthcare utilization for the final algorithm training. The remaining $435 - n$ labels were used to evaluate the out of sample prediction performance of the trained algorithms. We repeated this process 100 times and averaged over

TABLE 3 Model Sizes and AUCs for final algorithm training with 100 or 200 labeled samples (AUC_{100}, AUC_{200}) for the strong signal when $\mathbf{S} \not\perp\!\!\!\perp \mathbf{X} \mid Y$ based on (i) supervised training with all features and 100 or 200 labeled samples (L_{100}, L_{200}) or with features selected via the two-step approach ($\text{L}_{100}^{\text{2step}}$, $\text{L}_{200}^{\text{2step}}$) (ii) the automated feature selection method with and without resampling (AutoClust, AutoClust_R), (iii) a penalized regression with $S_1 + S_2$ or $\mathbf{S}$ as the outcome ($\text{PenReg}_{S_1+S_2}$, $\text{PenReg}_{\mathbf{S}}$), (iii) the rank correlation method based on $S_1 + S_2$ ($\text{RCor}_{S_1+S_2}$) (iv) the extreme sampling method (Extreme) and (iv) the method of Agarwal et al. (2016) based on $S_1 + S_2 \geq 1$ as the silver standard label ($\text{Agarwal}_{S_1+S_2\geq 1}$).

Method	Model Size	AUC_{100}	AUC_{200}
(a) $p = 50$			
L_{100}	3.3	81.2	
$\text{L}_{100}^{\text{2step}}$	3.6	81.6	
L_{200}	6.8		85.8
$\text{L}_{200}^{\text{2step}}$	7.9		84.5
AutoClust	5.7	85.3	86.5
AutoClust_R	4	85.6	86.6
$\text{PenReg}_{S_1+S_2}$	30.6	83	86.1
$\text{PenReg}_{\mathbf{S}}$	19	83.9	86.1
$\text{RankCor}_{S_1+S_2}$	2.6	83.6	84.5
Extreme	2.3	82.1	83
$\text{Agarwal}_{S_1+S_2\geq 1}$	9.1	84.9	86.4
(b) $p = 100$			
Method	Model Size	AUC_{100}	AUC_{200}
L_{100}	1.1	76.9	
$\text{L}_{100}^{\text{2step}}$	0.8	81.2	
L_{200}	5.1		84.3
$\text{L}_{200}^{\text{2step}}$	5.7		84.1
AutoClust	7.4	85.1	86.4
AutoClust_R	4.2	85.6	86.6
$\text{PenReg}_{S_1+S_2}$	53.1	80.6	85.2
$\text{PenReg}_{\mathbf{S}}$	21.2	83.8	86
$\text{RankCor}_{S_1+S_2}$	2.6	83.6	84.4
Extreme	2	81.3	82.1
$\text{Agarwal}_{S_1+S_2\geq 1}$	12.6	84.6	86.1

the trials. For clarity of presentation, we compare the performance of our method to models trained with other existing unsupervised feature selection methods as discussed in the Simulation section.

In Table 4, we present the coefficient estimates for the features that are selected in at least 50% of the trials, the corresponding AUC and the average number of features selected when feature selection is performed via different methods. Our AutoClust and AutoClust_R methods reduced the set of 79 candidate features to 33 and 27, respectively. The $\text{RankCor}_{S_1+S_2}$ and $\text{Agarwal}_{S_1+S_2\geq 1}$ methods selected significantly more features while the Extreme method selected 24 features. When using only 100 labels, all of trained algorithms placed the majority of the weights on the two surrogate variables. However, when training with features selected by AutoClust, AutoClust_R, or the extreme sampling method the weights are more spread out on other features. With 100 labels, the mean AUC of the algorithm trained with all features is 0.912 while algorithms trained with features selected by AutoClust or AutoClust_R attained an AUC of 0.927 and 0.928 respectively. The $\text{RankCor}_{S_1+S_2}$ and $\text{Agarwal}_{S_1+S_2\geq 1}$ method performed similarly to supervised training while the Extreme method performed well in this example, attaining an AUC of 0.934. When 200 labels are available for training, more features were selected in the final algorithm across all methods and the algorithms generally performed better. The algorithm trained with all features and 200 labels attained an AUC of 0.928 while the AUC of the algorithm trained with features selected by AutoClust_R and the Extreme method is around 0.945 $\sim$ 0.948. Although the extreme method performs comparably to the AutoClust_R method in this RA study, we find that our AutoClust_R attains more robust performance across different settings as demonstrated in the simulation studies. Overall, if we train algorithms using features selected via AutoClust_R, we require about half of the labels as those required by the standard supervised procedure to achieve the same accuracy. This again illustrates the potential of the proposed AutoClust_R methods in reducing the gold standard labeling effort and in turn allowing for the algorithm to be developed in a shorter time frame.

4 | DISCUSSION

In the context of EHR phenotyping, we have presented an automated feature selection method utilizing a two-step procedure involving a first step of model-based clustering followed by a second step of regularized regression on "silver standard labels" obtained from the first step. Our method makes efficient use of fully observed surrogate features that are moderately predictive of the true phenotype and relate to other candidate features only through the true phenotype status. This allows us to turn surrogates $\mathbf{S}$ into $\pi_{\mathbf{S}} = P(Y = 1 \mid \mathbf{S})$ as silver standard labels and subsequently assess the predictiveness of $\mathbf{X}$ for Y based on the predictiveness of $\mathbf{X}$ for $\pi_{\mathbf{S}}$. This approach advances previous work by accommodating non-binary and/or multiple surrogates and potentially high-dimensional predictors. Although our approach involves mixture model fitting which relates to the latent class analysis (LCA), for example, (Skrondal and Rabe-Hesketh, 2010; Vermunt, 2010) our goal and approach are substantially different from the LCA literature in that we aim to develop a robust feature selection procedure under possible mis-specifications of the fitted models.

Under a correct specification of a parametric mixture model for $\mathbf{S} \mid Y$, it may be feasible to estimate all model parameters simultaneously via an EM algorithm. However, when the mixture model is mis-specified, it is unclear whether the resulting estimator for $\boldsymbol{\beta}_0$ remains consistent up to a scalar multiplier. On the other hand, our procedure is valid regardless of the adequacy of the mixture model provided that $\pi_{\mathbf{S}}$

TABLE 4 (a) Number of features included in the algorithm training as well as coefficient estimates and AUC of the resulting algorithms trained with (b) $n = 100$ and (c) $n = 200$ labels when fitting the full model (L) as well as when only including features selected based on the AutoClust, AutoClust$_R$, RankCor$_{S_1+S_2}$, Extreme, and Agarwal$_{S_1+S_2 \geq 1}$ methods for the EMR-based study of RA.

		(a) Number of features selected				
		AutoClust	AutoClust$_R$	RankCor$_{S_1+S_2}$	Extreme	Agarwal$_{S_1+S_2 \geq 1}$
		33.00	26.80	69.20	23.80	45.10
(b) Coefficient and AUC estimates with $n = 100$ labels						
	L$_{100}$	AutoClust	AutoClust$_R$	RankCor$_{S_1+S_2}$	Extreme	Agarwal
RA$_{ICD}$	0.35	0.66	0.7	0.4	0.73	0.58
RA$_{NLP}$	0.27	0.68	0.74	0.32	0.85	0.49
Methotraxate$_{NLP}$	0.08	0.14	0.14	0.09	0.13	0.11
AM Stiffness$_{NLP}$	0.04	0.12	0.13	0.05	0.15	
Echography$_{NLP}$					−0.21	
AUC	91.1	92.7	92.8	91.2	93.4	91.7
(c) Coefficient and AUC estimates with $n = 200$ labels						
	L$_{200}$	AutoClust	AutoClust$_R$	RankCor$_{S_1+S_2}$	Extreme	Agarwal
RA$_{ICD}$	0.7	0.85	0.87	0.72	0.87	0.79
RA$_{NLP}$	0.72	1.26	1.35	0.79	1.52	1.07
Methotraxate$_{NLP}$	0.13	0.15	0.15	0.14	0.14	0.15
MRI$_{NLP}$	−0.05			−0.09		
AM Stiffness$_{NLP}$	0.1	0.28	0.29	0.12	0.29	0.17
Physiotherapy$_{NLP}$	−0.07					−0.15
Echography$_{NLP}$	−0.08				−0.53	−0.26
Note Count$_{NLP}$		−0.37	−0.41		−0.36	
Antinuclear Antibodies$_{NLP}$		−0.11	−0.13			
Redness$_{NLP}$			−0.16	−0.06		
Intravenous Infusion$_{NLP}$					−0.15	
AUC	92.8	94.3	94.5	92.9	94.8	94.2

is predictive of Y and $\mathbf{S} \perp \mathbf{X}$ given Y. It is thus feasible to construct $\pi_{\mathbf{S}}$ using alternative summaries of $\mathbf{S}$ or more stable clustering algorithms. For example, one may impose naive Bayes working assumption that $S_1, \ldots S_k$ are independent of each other given Y, or in other words that the covariance matrix of $\mathbf{S}|Y$ is diagonal. When $\mathbf{S}$ consists of binary and/or count data, binomial or Poisson mixture models can be used. While many choices of $\pi_{\mathbf{S}}$ are valid for the feature selection, the $\pi_{\mathbf{S}}$ that best approximates $P(Y = 1 \mid \mathbf{S})$ provides the closest approximation to the full likelihood with the observed Y. This choice yields the most efficient estimation and resulting feature selection as demonstrated in our numerical results. Additionally, it is worth noting that $\mathbf{S}$ is generally selected to contain the strongest predictors of Y. Several previous studies have found that the main ICD code and the number of NLP mentions for the phenotype are the most influential predictors of disease (Liao, et al., 2010; Liao et al., 2015; Xia et al., 2013; Yu et al., 2016). We thus suggest using these two features for $\mathbf{S}$ in the EHR setting while in other settings the choice of $\mathbf{S}$ may be similarly made based on domain knowledge.

Additionally, we do require the assumption of $E(\mathbf{b}^\mathsf{T}\mathbf{X} \mid \boldsymbol{\beta}_0^\mathsf{T}\mathbf{X})$ to be linear in $\boldsymbol{\beta}_0^\mathsf{T}\mathbf{X}$ and $\mathbf{S} \perp \mathbf{X}|Y$. However, our numerical studies suggest that the methods are not very sensitive to the departure of this assumption. Results from our simulation studies and real EHR data application also demonstrate the value of the unsupervised feature selection in reducing the number of labels needed to train an accurate and generalizable algorithm. As manual chart review is the rate-limiting step for developing a phenotyping algorithm, our procedure holds promise in improving efficiency by decreasing the burden of chart review. While our approach only utilized unlabeled data for feature selection, semi-supervised methods that combine both labeled and unlabeled data to estimate the model parameters would be valuable for EHR applications. A simple approach is to replace the quasi-binomial likelihood $\sum_{i=1}^{N} \ell(\boldsymbol{\theta}^\mathsf{T}\vec{\mathbf{X}}_i, \widehat{\pi}_{\mathbf{S}_i})$ by $\sum_{i=1}^{N} \ell(\boldsymbol{\theta}^\mathsf{T}\vec{\mathbf{X}}_i, \widehat{\pi}_{\mathbf{S}_i}) + \sum_{j=N+1}^{N+n} \ell(\boldsymbol{\theta}^\mathsf{T}\vec{\mathbf{X}}_i, Y_i)$. Robust yet efficient semi-supervised approaches for feature selection warrant further research.

We did not observe drastic differences between the AutoClust procedure and the AutoClust$_R$ procedure with respect to the accuracy in feature selection and prediction. However, when the size of the labeled set for downstream algorithm training is small, the AutoClust$_R$ procedure is generally better since it tends to select a smaller number of variables. In practice we find that $\rho_{cut} = 0.5$ works well, but a data-driven approach to select the cut-off value may increase the robustness of the procedure. One possibility is to estimate the threshold by carrying out the procedures on random permuted data under which $\pi_{\mathbf{S}}$ and $\mathbf{X}$ are independent. Lastly, when the EHR data size is not too large, one may replace subsampling by standard resampling such as the bootstrap or wild bootstrap.

Our proposed AutoClust and AutoClust$_R$ feature selection methods also have the potential of leading to unsupervised learning with additional steps. For example, one may potentially perform unsupervised classification based on $\mathbf{S}$ and $\widehat{\boldsymbol{\beta}}^{\mathsf{T}}\mathbf{X}$ via unsupervised clustering or simple procedures such as the k-means. In our RA example, when performing a three dimensional clustering using a normal mixture approach, the unsupervised algorithm yields an AUC of 0.908. More robust and accurate unsupervised phenotyping methods warrant further research.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.
Web Appendices, Tables, and Figures referenced in Sections 2.1 and 3.1 are available with this paper at the Biometrics website on Wiley Online Library. The corresponding R code for the proposed procedures is also available at the Biometrics website on Wiley Online Library.

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